Linking Wastewater-Based Surveillance with Hospital Infections in a Northern California Community Hospital, November 2024 to June 2025

Mingjun Jiang[2] , Erika Viana Cardenas[1], Guillermo Rodriguez Nava[1], Alessandro Zulli[2], Sehee Jong[2], Wajeeha Tariq[1], Eugenia Miranti[1], Mindy Sampson[1], Nida Subhani[3] Alexandria Boehm[2], Jorge L. Salinas[1]

[1]Division of Infectious Diseases & Geographic Medicine, Stanford University School of Medicine, [2]Department Civil and Environmental Engineering, Stanford University, [3]Stanford Health Care Tri-Valley

Background

Antimicrobial resistance (AMR) is an escalating global public-health threat[1].Wastewater-based surveillance (WBS) enables population-level detection of pathogens and antimicrobial resistance genes (ARGs) shed by humans without individual testing.

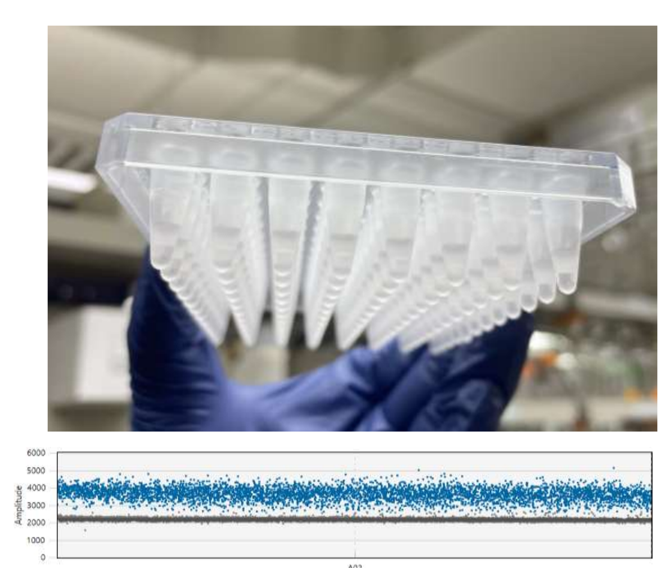
In this study, we monitored clinically relevant ARGs in hospital wastewater and evaluated their associations with hospital-reported infections to inform infection prevention and control (IPC). We further evaluated the relationships between the hospital ARGs to reveal co-occurrence patterns.

Methods

Step 1
Weekly composite samples using autosampler from hospital wastewater pipe

Step 2
Spined down solids and Nucleic Acids Extraction

Step 3
Droplet digital PCR for Quantification



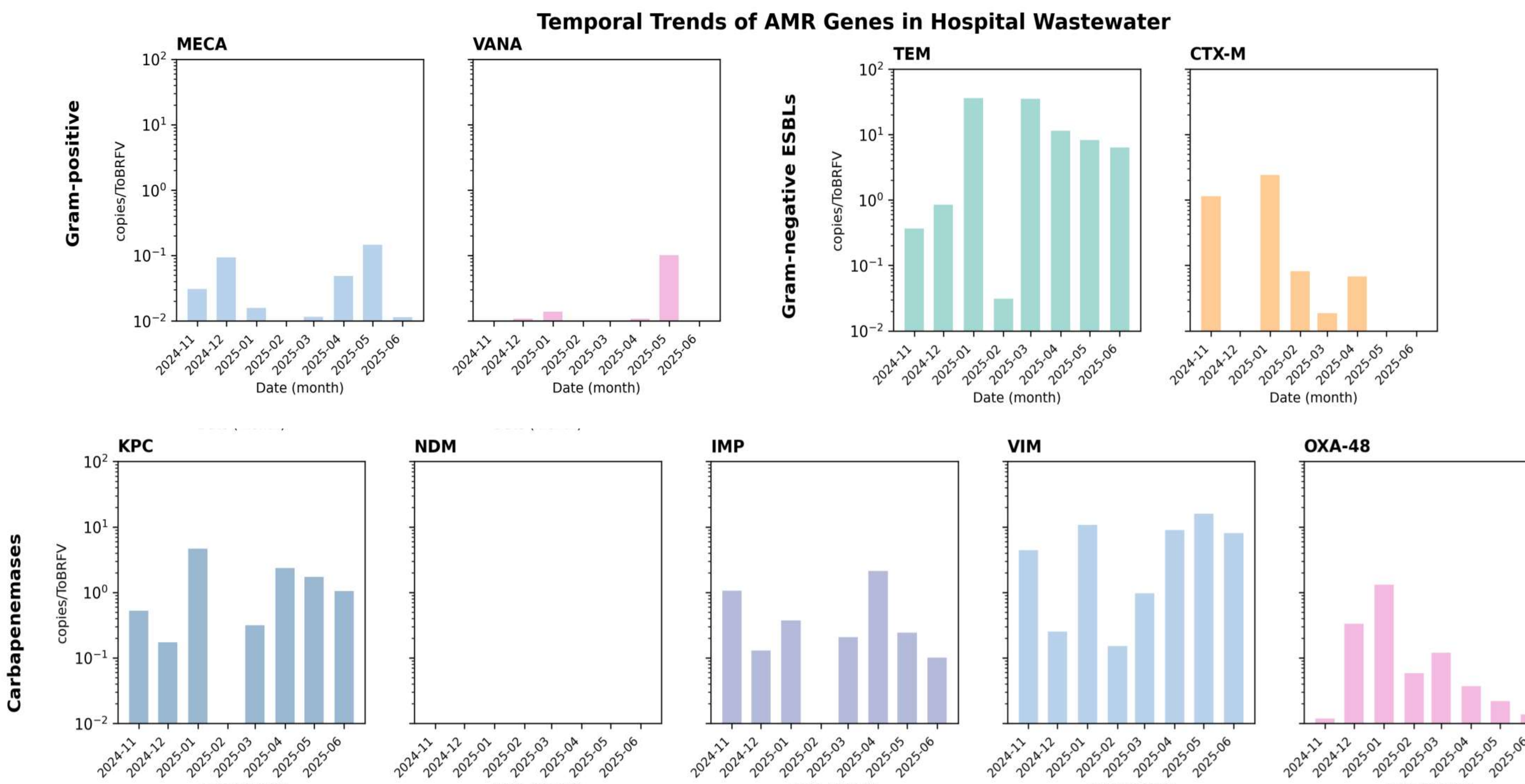
Results

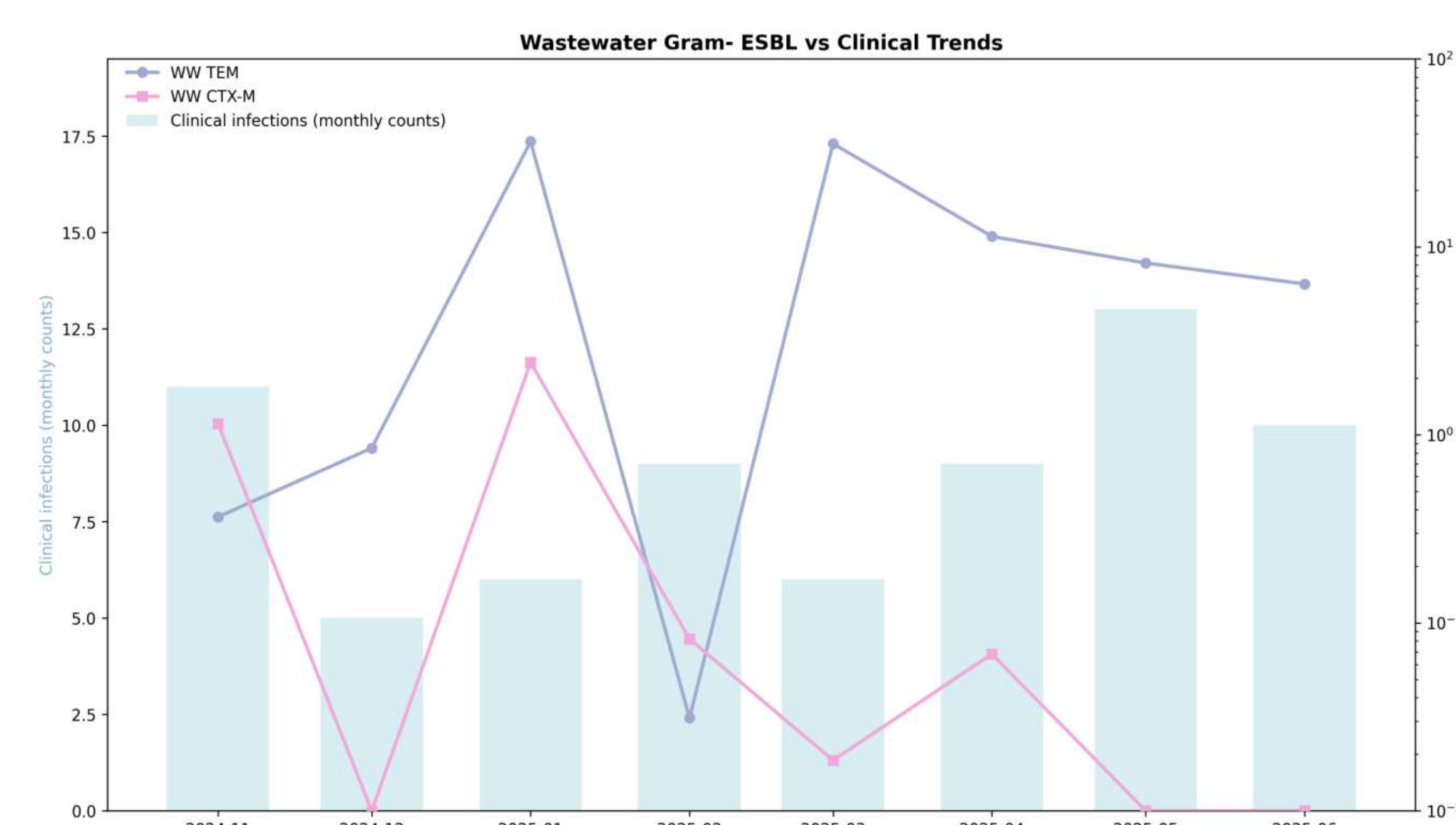
Clinically relevant ARGs were detected in wastewater samples. A total of nine ARG targets were quantified, including Gram-positive resistance genes (mecA and vanA), Gram-negative extended-spectrum β-lactamase genes (TEM and CTX-M), and carbapenemase genes (KPC, NDM, IMP, VIM, and OXA-48). ARG concentrations were normalized to Tomato brown rugose fruit virus (ToBRFV), an alternative viral fecal marker to pepper mild mottle virus (PMMoV) that is more ubiquitously detected in environmental wastewater Samples[2].

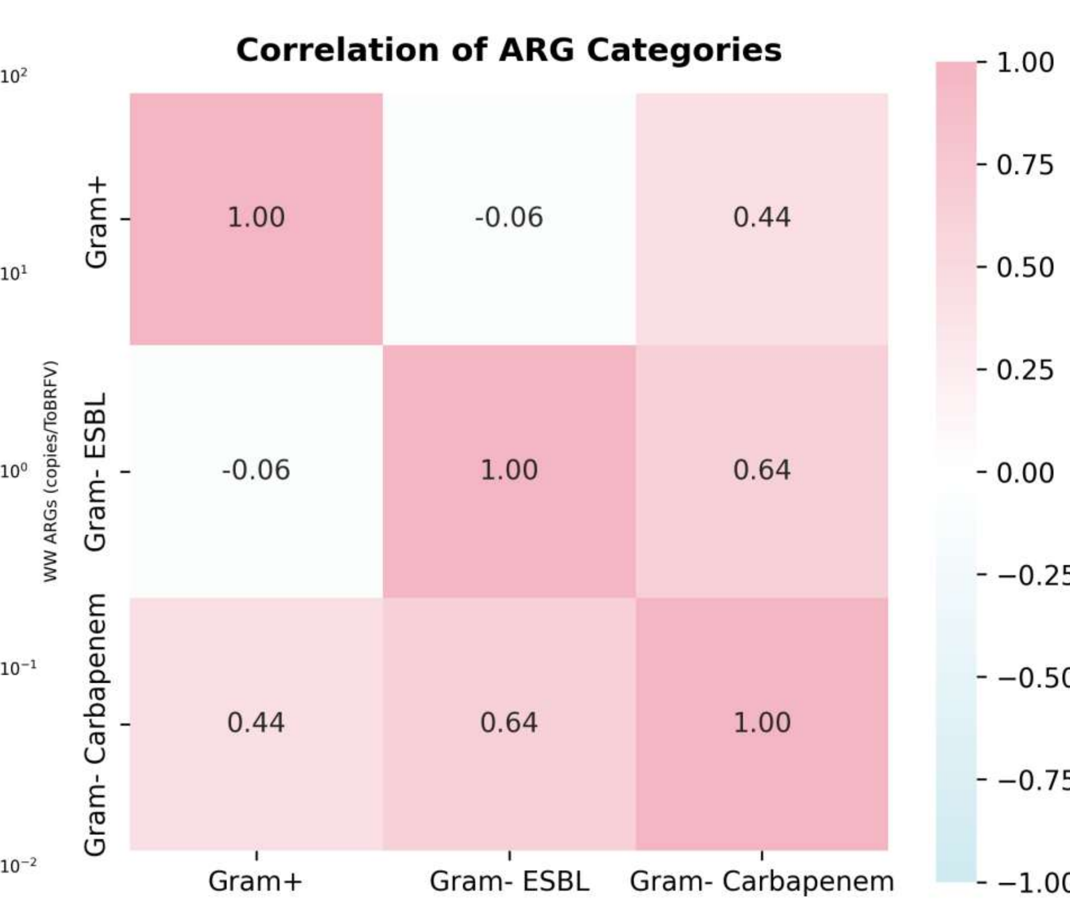
The ARGs TEM, VIM and KPC were found in high concentration. ARGs also exhibited potential seasonal patterns. Gram-negatives extended-spectrum beta-lactamases showed similar trend with hospital infections

Stronger correlations were observed between the two Gram- resistance categories, whereas Gram- carbapenemase genes exhibited moderate positive correlations with Gram+ resistance genes.

Results







Conclusions

Wastewater-based surveillance detected clinically relevant ARGs, with TEM, VIM, and KPC observed at high concentrations. Gram- ESBL genes showed temporal patterns aligned with hospital infections. Stronger correlations were observed within two Gram- resistance categories and moderate positive correlations were found between Gram- carbapenemase and Gram+ categories. These findings highlight the utility of wastewater surveillance for supporting infection prevention and control.

[1] GBD 2021 Antimicrobial Resistance Collaborators (2024). *The Lancet* 404:1199–1226. https://doi.org/10.1016/S0140-6736(24)01867-1
[2] Natarajan *et al.* (2023). *Applied and Environmental Microbiology* 89:e00583-23. https://doi.org/10.1128/aem.00583-23.